TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2024/2025

JANUARY EXAMINATION

AMMS1623 CALCULUS AND ALGEBRA

TUESDAY, 7 JANUARY 2025 | TIME: 2.00 PM – 4.00 PM (2 HOURS)

DIPLOMA IN COMPUTER SCIENCE

DIPLOMA IN INFORMATION TECHNOLOGY

DIPLOMA IN SOFTWARE ENGINEERING

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

Question 1

a) Find the equation of the straight line that passes through the point (2, –3) and perpendicular to the line, $2x + 3y = 5$. (5 marks)

b) Find the shortest distance between the point (–2, 3) and the straight line $2x + 5y - 7 = 0$. (5 marks)

c) Find the centre and radius of a circle with equation

$$x^2 + y^2 + 4x - 2y - 20 = 0.$$ (5 marks)

d) Given the vectors $\mathbf{u} = \langle 2, 5\rangle$ and $\mathbf{v} = \langle -3, 3\rangle$. Find

(i) the magnitudes of vectors **u and v**, (1 mark)

(ii) the dot product $\mathbf{u} \cdot \mathbf{v}$, (2 marks)

(iii) the angle between **u** and **v**. (2 marks)

e) Given the vectors $\mathbf{u} = 3\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$ and $\mathbf{v} = \mathbf{i} + 2\mathbf{j} - 5\mathbf{k}$. Find a vector that is orthogonal to **u** and **v**. (5 marks)

[Total: 25 marks]

Question 2

a) Find the greatest common divisor (GCD) of 77 and 1260 by using the Euclidean algorithm and rewrite the GCD in the form of $77s + 1260t$, $s, t \in \mathbb{Z}$. Hence find the least common multiple of 77 and 1260. (7 marks)

b) In the context of $\mathbb{Z}_{11}$, calculate

(i) $5 \otimes 8$, (2 marks)

(ii) $6 \ominus 7$, (2 marks)

(iii) $8 \oplus 3$, (2 marks)

c) Solve the system of equations given below:

$x \equiv 3 \pmod 5$,
$x \equiv 5 \pmod 9$. (6 marks)

d) Differentiate each of the following functions with respect to x.

(i) $y = 3x^4 + 2 - \frac{1}{x}$ (3 marks)

(ii) $y = 2(3x^4 - 2)^5$ (3 marks)

[Total: 25 marks]

Question 3

a) Find the equation of the tangent line to the curve $y = \frac{1}{x+1}$ at the point (1, 2). (5 marks)

b) Find all the relative extreme points and inflection point of the curve $y = x^3 + 2x^2 - 4x + 1$. (7 marks)

c) Find the linear approximation of the function $f(x) = \sqrt{8 + x}$ at $x = 1$ and use it to approximate the numbers $\sqrt{8.88}$ and $\sqrt{9.08}$. (7 marks)

d) Suppose that an object is moving in a straight line and at time t second, the position of the object from the starting point in cm is given by $s(t) = -t^2 + 3t - 2$. Find

(i) the position of the object when it stops for a moment, (2 marks)

(ii) the initial velocity of the object, (2 marks)

(iii) the constant acceleration of the object. (2 marks)

[Total: 25 marks]

Question 4

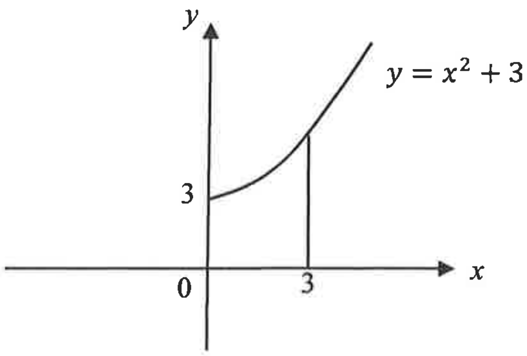
a) Find the indefinite integral for each of the following:

(i) $\int (3x + 1)^4 dx.$ (3 marks)

(ii) $\int 6x(3x^2 + 1)^3 dx.$ (6 marks)

b) Find the average value of $f(x) = 4x + 3$ over the interval from $x = 0$ to $x = 3$. (5 marks)

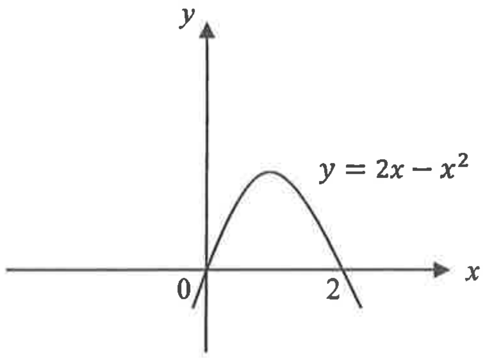
c) The diagram given below shows part of the curve $y = x^2 + 3$ and the straight-line $x = 3$.



Calculate the area of the region bounded by the curve $y = x^2 + 3$, the straight-line $x = 3$, the x-axis and the y-axis. (5 marks)

Question 4 (Continued)

d) The diagram given below shows the curve $y = 2x - x^2$.



If the region bounded by the curve $y = 2x - x^2$ and the x-axis is revolved through 360° about the x-axis, find the volume of the solid generated, in terms of π. (6 marks)

[Total: 25 marks]

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MATHEMATICAL FORMULAE

Distance Formula

The **distance** between the points $A(x_1, y_1)$ and $B(x_2, y_2)$ in the plane is

$$d(A, B) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Midpoint Formula

The **midpoint** of the line segment from $A(x_1, y_1)$ to $B(x_2, y_2)$ is

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

Distance between a Point and a Line

The **distance** d from a point $Q(x_0, y_0)$ to the line $ax + by + c = 0$ is

$$d = \frac{|a(x_0) + b(y_0) + c|}{\sqrt{a^2 + b^2}}$$

Equation of a Circle

An equation of the circle with **centre** (h, k) and **radius** r is

$$(x - h)^2 + (y - k)^2 = r^2$$

Vectors in Two Dimensions

If $\mathbf{u} = \langle a_1, a_2 \rangle$ and $\mathbf{v} = \langle b_1, b_2 \rangle$, then their **dot product** is

$$\mathbf{u} \cdot \mathbf{v} = a_1 b_1 + a_2 b_2$$

If θ is the **angle** between $\mathbf{u}$ and $\mathbf{v}$, then

$$\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}| \cos\theta$$

The **angle** θ between $\mathbf{u}$ and $\mathbf{v}$ satisfies

$$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$$

Vectors in Three Dimensions

The **distance** between the points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ is given by the

$$d(P, Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

If $\mathbf{u} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{v} = \langle b_1, b_2, b_3 \rangle$ are vectors in space, then their **dot product** is

$$\mathbf{u} \cdot \mathbf{v} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

The Cross Product of Vectors in Space

If $\mathbf{u} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{v} = \langle b_1, b_2, b_3 \rangle$ are vectors in space, then their **cross product** is the vector

$$\mathbf{u} \times \mathbf{v} = (a_2 b_3 - a_3 b_2)\mathbf{i} - (a_1 b_3 - a_3 b_1)\mathbf{j} + (a_1 b_2 - a_2 b_1)\mathbf{k}$$

We can calculate the **cross product** using **determinant.**

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

The scalar triple product

The product $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$ is called the **scalar triple product** of the vectors $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$. You can check that the scalar triple product can be written as the following determinant:

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Equations of Lines and Planes

A **line** passing through the point $P(x_0, y_0, z_0)$ and parallel to the vector $\mathbf{v} = \langle a, b, c \rangle$ is described by the **parametric equations**

$$x = x_0 + at, y = y_0 + bt, z = z_0 + ct$$

where t is any real number.

A **plane** containing the point $P(x_0, y_0, z_0)$ and having the normal vector $\mathbf{n} = \langle a, b, c \rangle$ is described by the **plane equation**

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

The Distance from a Point to a Line

Distance from a Point S to a Line Through P Parallel to v

$$d = \frac{|\overrightarrow{PS} \times \mathbf{v}|}{|\mathbf{v}|}$$

The Distance from a Point to a Plane

If P is a point on a plane with normal $\mathbf{n}$, then the distance from any point S to the plane is

$$d = \left| \overrightarrow{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right|$$

where $\mathbf{n} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$ is normal to the plane.

Product Rule

$$\frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + g(x)f'(x)$$

Quotient Rule

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$$

The Chain Rule

To differentiate $f(g(x))$, differentiate first the outside function $f(x)$ and substitute $g(x)$ for x in the result. Then, multiply by the derivative of the inside function $g(x)$. Symbolically,

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x).$$

Differentiation Formula	**Corresponding Integration Formula**
$\frac{d}{dx}(x^r) = rx^{r-1}$	$\int rx^{r-1}dx = x^r + C$ or
$\frac{d}{dx}\left(\frac{x^{r+1}}{r+1}\right) = x^r$	$\int x^r dx = \frac{x^{r+1}}{r+1} + C, r \neq -1$
$\frac{d}{dx}(e^x) = e^x$	$\int e^x dx = e^x + C$
$\frac{d}{dx}(\ln\lvert x\rvert) = \frac{1}{x}$	$\int \frac{1}{x}dx = \ln\lvert x\rvert + C$
$\frac{d}{dx}(\sin x) = \cos x$	$\int \cos x\, dx = \sin x + C$
$\frac{d}{dx}(\cos x) = -\sin x$	$\int \sin x\, dx = -\cos x + C$
$\frac{d}{dx}(\tan x) = \sec^2 x$	$\int \sec^2 x\, dx = \tan x + C$

Integration by Substitution

$$\int f(g(x))g'(x)dx = F(g(x)) + C$$

Integration by Parts

$$\int f(x)g(x)dx = f(x)G(x) - \int f'(x)G(x)dx$$

Linear Approximation

$$f(x) \approx L(x) = f(a) + f'(a)(x - a)$$

The mean value of $f(x)$ over the interval from $x = a$ to $x = b$.

$$\frac{1}{b-a}\int_a^b f(x)dx$$

The volume of the solid of revolution generated by revolving the region under $y = g(x)$ from $x = a$ to $x = b$ about the x-axis.

$$\int_a^b \pi[g(x)]^2 dx$$